

Heterogeneous Treatment Effects

CML Week 5

2026-09-29

Schedule

1. Design-based causal inference & Monte Carlo simulation
2. Covariate adjustment
3. Balancing weights
4. Double robustness, Double ML and TMLE
5. **Heterogeneous Treatment Effects**
6. HTE II: Neural Networks
7. Off-policy evaluation and optimization
8. Experimental design
9. Panel data and the new diff-in-diff
10. Partial Identification
11. Response adaptation
12. Interference

Thus far

- › Every week so far has targeted **one number**: the **average treatment effect** $\tau = E[Y(1) - Y(0)]$
- › Design-based inference: randomisation identifies it without a model
- › Covariate adjustment, balancing weights, double robustness: three routes to a *better estimate of the same average*



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- › *Medical treatments*: some patients benefit, others don't (or are harmed)
- › *Policy interventions*: context matters for effectiveness
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Goal today: understand why estimating heterogeneous effects is **fundamentally harder than it seems**, and where the ideas for doing it better come from.



(Holland 1986)

The Fundamental Problem of Causal Inference

For any individual unit i , we can never observe both $Y_i(1)$ and $Y_i(0)$ simultaneously.

- › Each unit has two potential outcomes: $Y_i(1)$ and $Y_i(0)$
- › We only observe: $Y_i = A_i Y_i(1) + (1 - A_i) Y_i(0)$
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Why it's fundamental:

- › Not a sample size problem (more data doesn't help)
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Implications:

- › We can make causal claims about populations or subgroups
- › We cannot make causal claims about specific individuals with certainty
- › All individual-level causal predictions are inherently model-dependent



What can we learn about individuals?

What we cannot identify:

- › Individual treatment effect: $\tau_i = Y_i(1) - Y_i(0)$
- › Whether a specific person benefited from treatment

What we can identify (under assumptions):

- › Average Treatment Effect: $\tau = E[Y(1) - Y(0)]$
- › Conditional Average Treatment Effect: $\tau(x) = E[Y(1) - Y(0) \mid X = x]$



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CATE as a compromise:

$$\tau(x) = E[Y(1) - Y(0) \mid X = x]$$

- › Not about individuals, but about groups defined by X
- › As X becomes more detailed, groups become smaller
- › But we still need multiple units with similar X to estimate $\tau(x)$
- › **Bias-variance tradeoff**: more granular X means smaller groups, higher variance



The role of randomization

For the ATE, randomization is powerful:

- › Ensures $A \perp (Y(1), Y(0))$
- › Simple difference in means is unbiased:
$$E[\bar{Y}_{A=1} - \bar{Y}_{A=0}] = \tau$$
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For HTEs, randomization helps less:

- › Still have $A \perp (Y(1), Y(0)) \mid X$
- › But now we need sufficient treated and control units at each value of X
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The curse of dimensionality:

- › With 10 binary covariates: $2^{10} = 1024$ subgroups
- › Even in a large RCT, many subgroups may have no treated or control units
- › Must **borrow strength across subgroups** (i.e., model)



How can we do machine learning?

The standard supervised learning paradigm:

1. Observe many examples of (X_i, Y_i)
2. Learn function $f : X \rightarrow Y$ that minimizes prediction error
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- › Input: covariates X
- › Output: treatment effect $\tau(x)$
- › Just use your favorite ML algorithm on observed (X_i, A_i, Y_i) !

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Why this fails for HTE:

- › **No observed target:** we never see $\tau_i = Y_i(1) - Y_i(0)$
- › **No natural loss function:** can't compute $(\hat{\tau}(X_i) - \tau_i)^2$ to optimize
- › **Validation is impossible:** can't check if predictions are good on a test set

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The problem with naive approaches

Consider the naive “difference in predictions” approach (**T-learner**):

1. Train $\hat{\mu}_1(x)$ to predict Y among treated units
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This conflates two distinct concepts:

- › **Prognostic variables**: X predicts Y regardless of treatment (baseline risk)
- › **Effect moderators**: X modifies the treatment effect



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The consequence: overfitting to noise in the heterogeneity, mistaking prognostic variables for moderators, and overconfident, miscalibrated predictions.

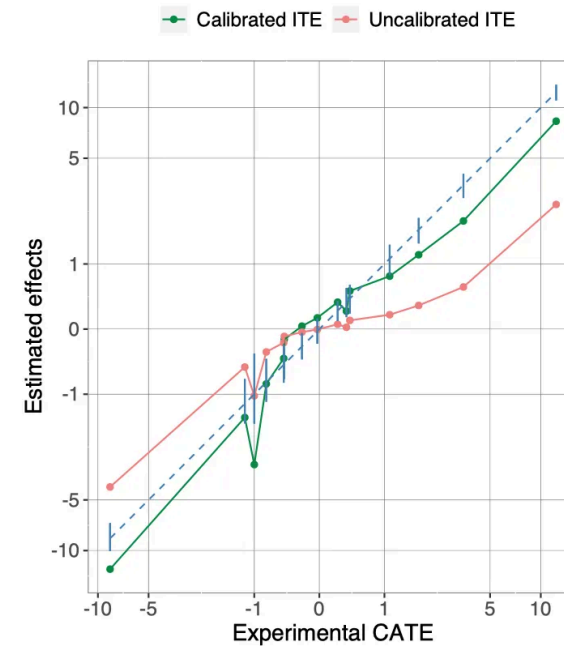
What we need: methods that explicitly target $\tau(x)$, not just Y



Bonus: (Leng and Dimmery 2024)

What can we observe about our HTE estimates?

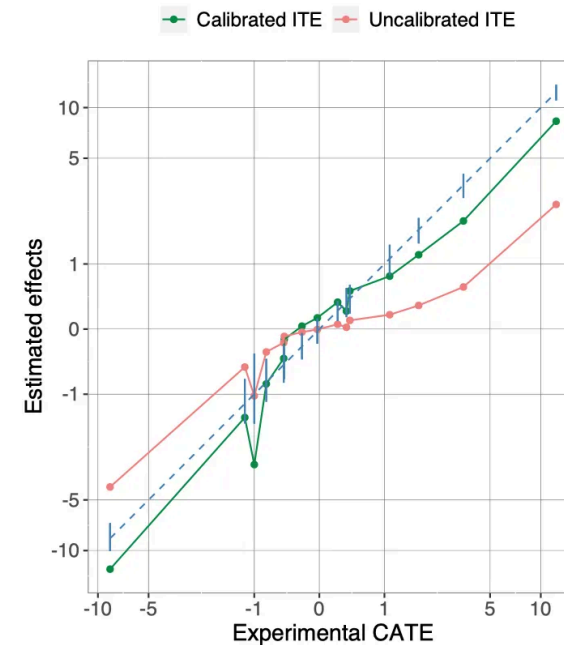
You ran an RCT, used a sophisticated HTE method and have $\hat{\tau}(X_i)$ for all units. **Are these estimates trustworthy?**



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Observation: we *can* observe subgroup average effects. For any subgroup S , the difference in means is unbiased for $E[\tau \mid X \in S]$. So build the subgroups from $\hat{\tau}(X_i)$ itself and check whether they deliver what they promise.



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Calibration check:

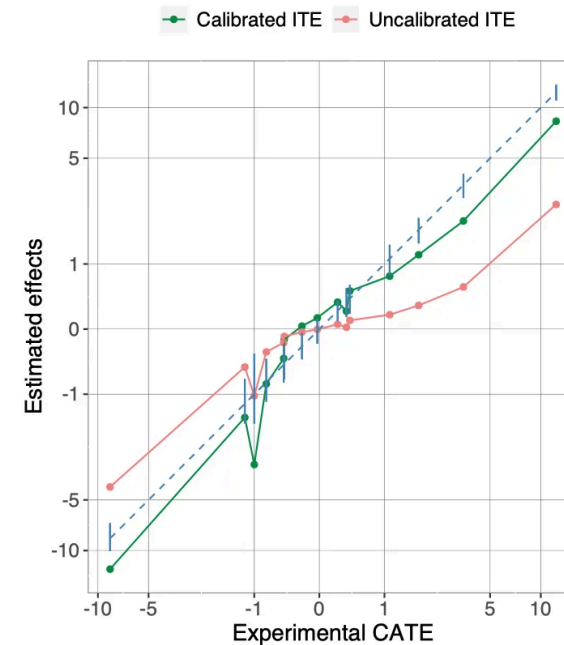
1. Sort units by $\hat{\tau}(X_i)$ into quantiles (e.g., deciles)

2. In each quantile q , compare the predicted effect

$$\begin{aligned}\bar{\hat{\tau}}_q &= \frac{1}{n_q} \sum_{i \in q} \hat{\tau}(X_i) \text{ with the observed one } \bar{\tau}_q \\ &= \bar{Y}_{A=1,q} - \bar{Y}_{A=0,q}\end{aligned}$$

3. Plot predicted vs. observed

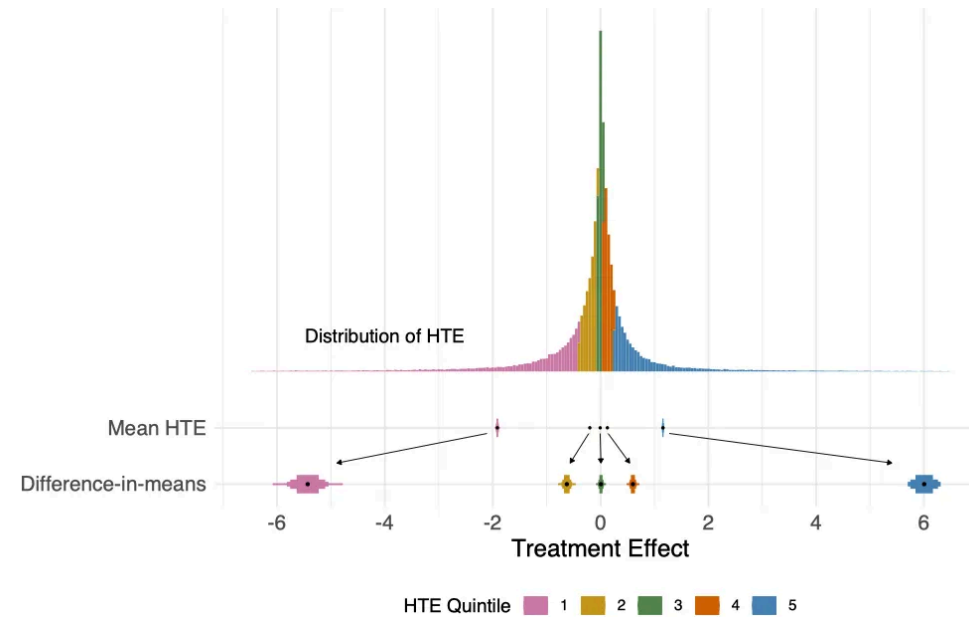
Perfect calibration: points lie on the 45-degree line



Calibration in practice

Evidence from large-scale experiments (Facebook, Criteo platforms):

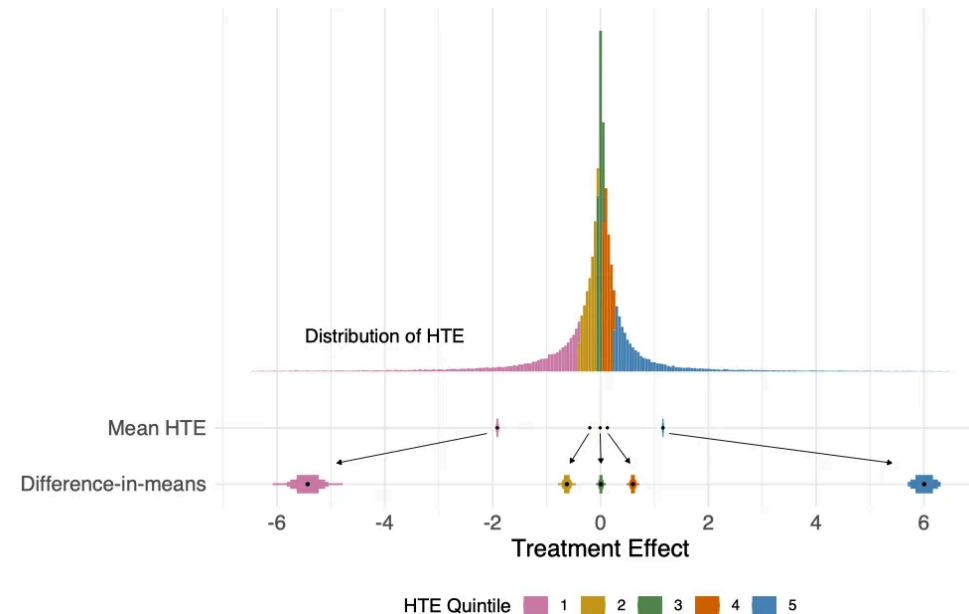
- › **Substantial miscalibration:** ML-based HTE estimates are often poorly calibrated
- › **Range compression:** predicted effects are tightly clustered, observed effects are much wider



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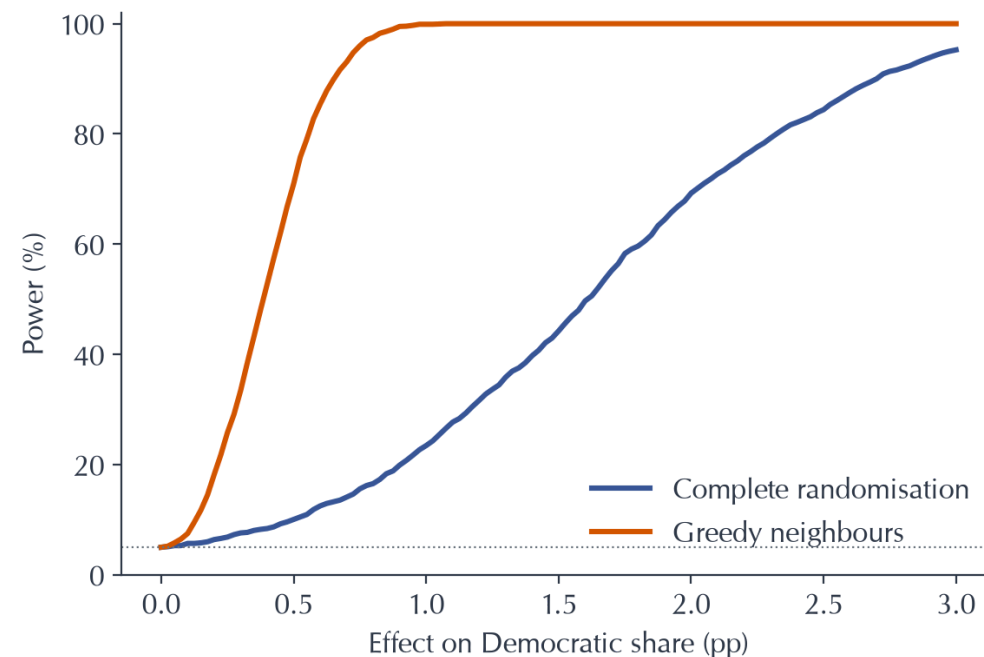
Why does this happen?

- › Optimization targets prediction of Y , not τ
- › Small treatment effects relative to outcome variance
- › Prognostic/predictive confusion amplified by complex models

Bonus: ([Arbour, Dimmery, and Rao 2021](#))

A complementary approach: can we do better if we *choose* how to assign treatment?

- › Not all units are equally informative about $\tau(x)$
 - » Units with common covariate values: less valuable (we already have many like them)
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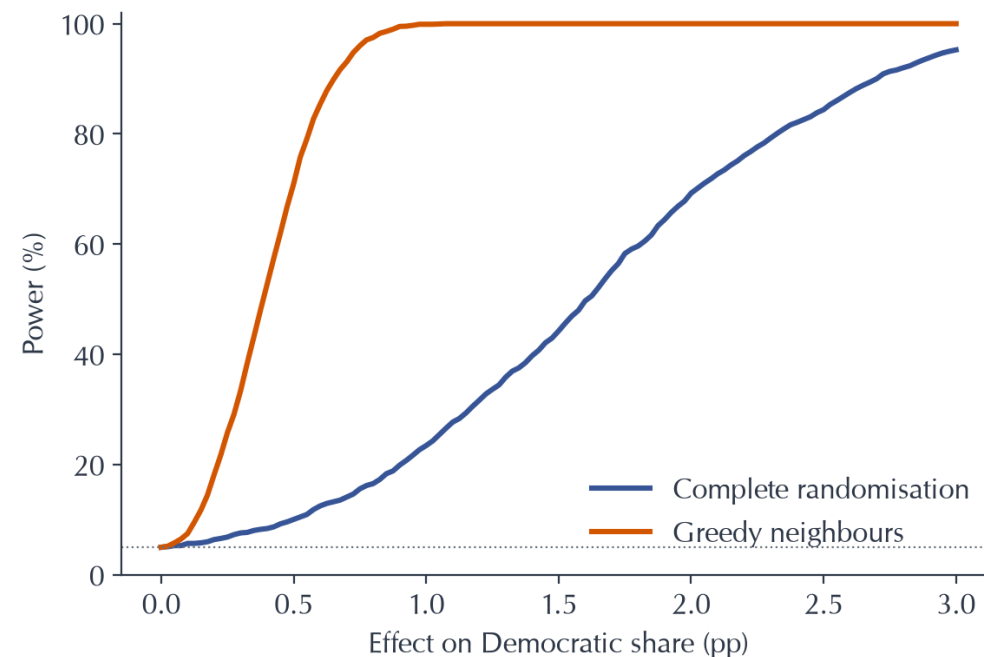
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Locally balanced assignments:

- › Ensure good balance not just on average, but locally
- › Assignments are well distributed across covariate space

Connection to today's themes:

- › Design can't solve the Fundamental Problem
- › But it can make sure you don't shoot yourself in the foot



Takeaways

- › **The Fundamental Problem matters:**
 - » We can never observe individual treatment effects
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 - » This makes HTE estimation fundamentally different from supervised learning
 - » Standard ML intuitions can mislead us
- › **What makes HTE hard:**
 - » No ground truth for validation
 - » Easy to overfit to noise
 - » Prognostic vs. predictive confusion
- › **Practical implications:**
 - » Need specialized methods (which students will present today)
 - » Need calibration checks when possible
 - » Skepticism is warranted for HTE claims

Open questions

1. **Adapting a workhorse:** Random forests predict Y well. Can a forest target $\tau(x)$ directly, and still deliver valid confidence intervals?
 - › *Motivates:* ([Wager and Athey 2018](#)) on causal forests
2. **Combining models:** If we already have good outcome models, how should we combine them to get $\tau(x)$, and when does each recipe win?
 - › *Motivates:* ([Künzel et al. 2019](#)) on meta-learners
3. **Double robustness for CATEs:** Last week's machinery hedged between two models for one number. Can it be pointed at $\tau(x)$, and how fast can we learn it?
 - › *Motivates:* ([Kennedy 2020](#)) on the DR-learner
4. **Inference without a target:** If τ_i is never observed, what *can* we test about heterogeneity in an RCT with honest inference?
 - › *Motivates:* ([Chernozhukov et al. 2025](#)) on generic ML inference



More to read

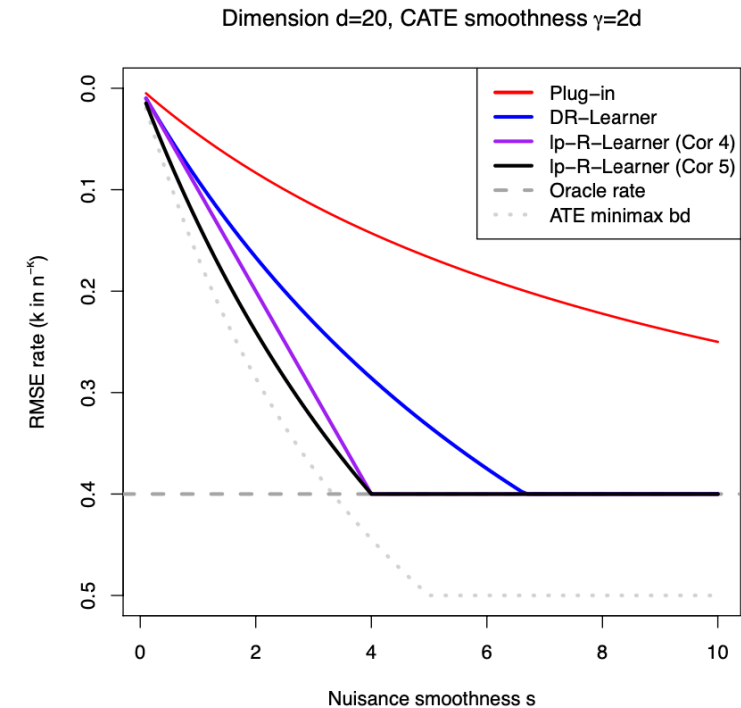
Paper	Why you'd read it
(Athey and Imbens 2019)	The review that frames today: the ML toolkit for applied economists, and where its causal uses part ways with prediction.
(Wager and Athey 2018)	Causal forests: <i>honest</i> trees (split on one half, estimate on the other) give consistency and asymptotic normality for $\tau(x)$.
(Künzel et al. 2019)	Meta-learners: S-, T- and X-learners as recipes for any base learner. The X-learner wins when the arms are unbalanced.
(Kennedy 2020)	The DR-learner: regress a doubly robust pseudo-outcome on X . The error is a product of nuisance errors, as last week.
(Chernozhukov et al. 2025)	Generic ML: test for heterogeneity (BLP), sort groups by it (GATES), describe them (CLAN), with valid inference after sample splitting.
(Athey and Imbens 2016) ; (Athey, Tibshirani, and Wager 2019)	The honest causal tree before causal forests, and the general framework (forests for any local moment condition) after.
(Nie and Wager 2021)	The R-learner: Robinson's residual-on-residual regression as a loss for $\tau(x)$, with quasi-oracle rates.



Deeper dive: (Kennedy 2020)

The DR-learner

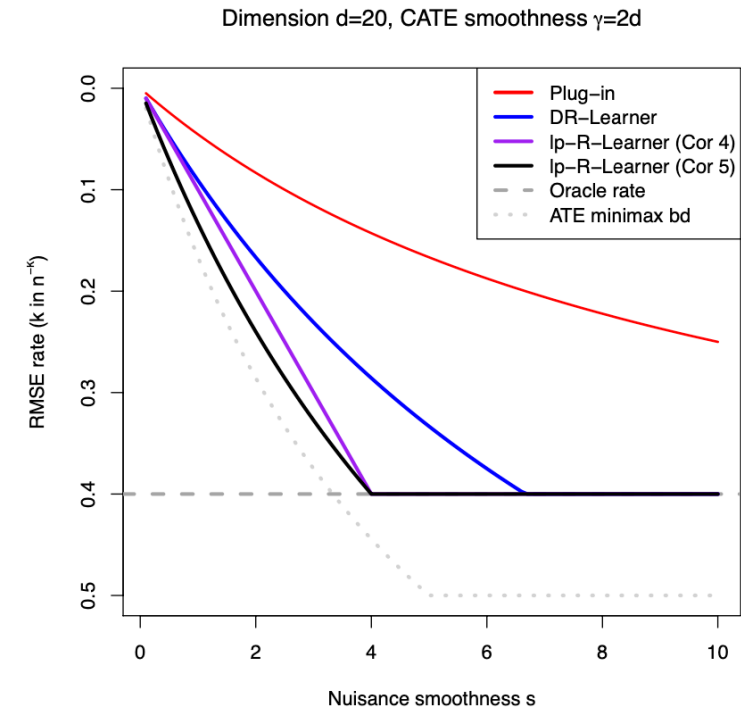
1. **Nuisance training:** the propensity score $\hat{e}$ (in an RCT, known) and the regression functions $\hat{\mu}_1, \hat{\mu}_0$



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2. **Pseudo-outcome:** an unbiased estimate of each unit's treatment effect

$$\hat{\varphi} = \frac{A - \hat{e}}{\hat{e}(1 - \hat{e})} (Y - \hat{\mu}_A) + \hat{\mu}_1 - \hat{\mu}_0$$

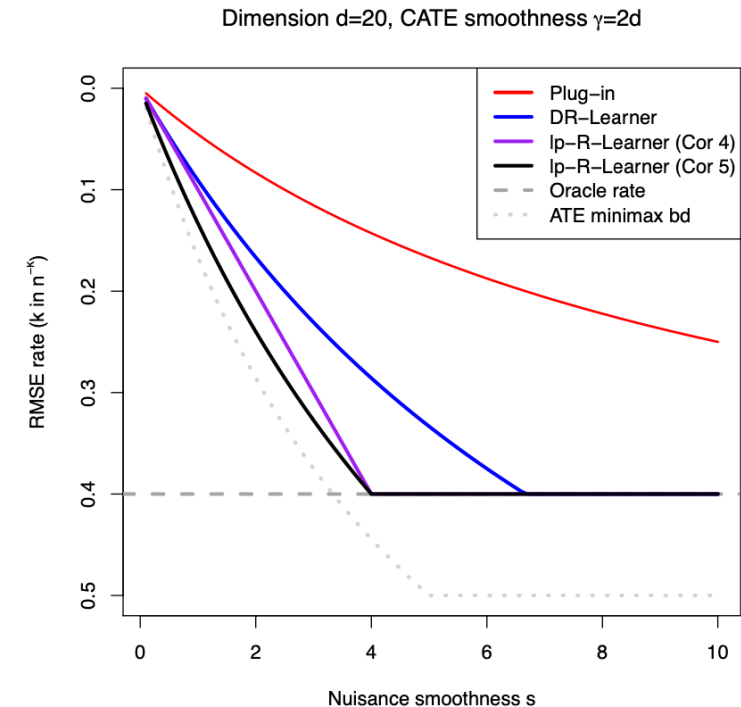


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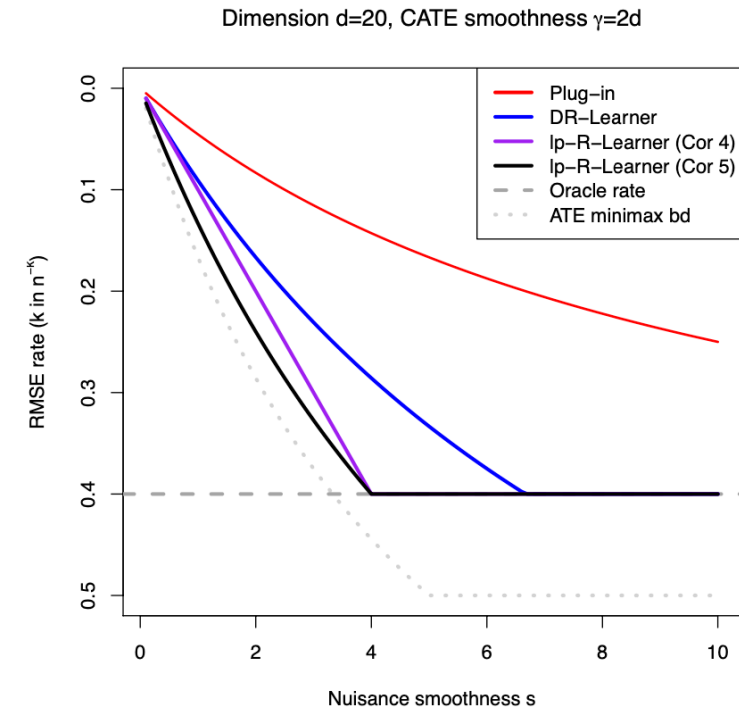


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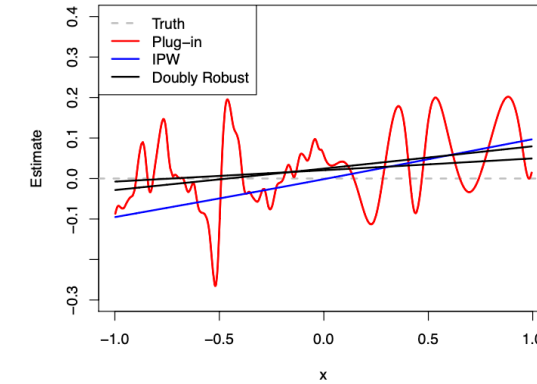
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3. **Smooth the pseudo-outcome:** regress $\hat{\varphi}$ on X with any ML method. The fit is $\hat{\tau}_{\text{dr}}(\mathbf{x})$.
- › Nuisance training and smoothing must be on separate subsamples (**cross-fitting**); rotate the folds and average.



Why does this work?

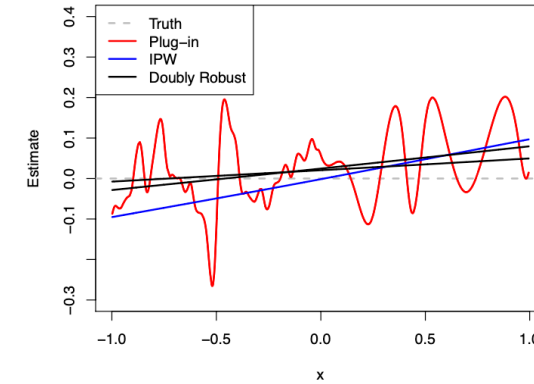
- › φ is the uncentred **efficient influence function** of the ATE: AIPW, pointwise
- › Its conditional mean is the CATE: $E[\varphi | X] = \tau(X)$. Smoothing φ over X is a regression with the right target.
- › Plugging in $\hat{e}, \hat{\mu}_a$ adds bias only through the **product of their errors**: doubly robust at every x .



Kennedy's simulation: rough μ_a , flat τ .
Only the plug-in inherits the roughness.

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- › Plugging in $\hat{e}, \hat{\mu}_a$ adds bias only through the **product of their errors**: doubly robust at every x .
- › Error = *oracle* error of regressing the true φ on X , plus $\hat{b}(x)^2$
- › Harder than the ATE by a factor growing in $\dim X$, shrinking in τ 's smoothness

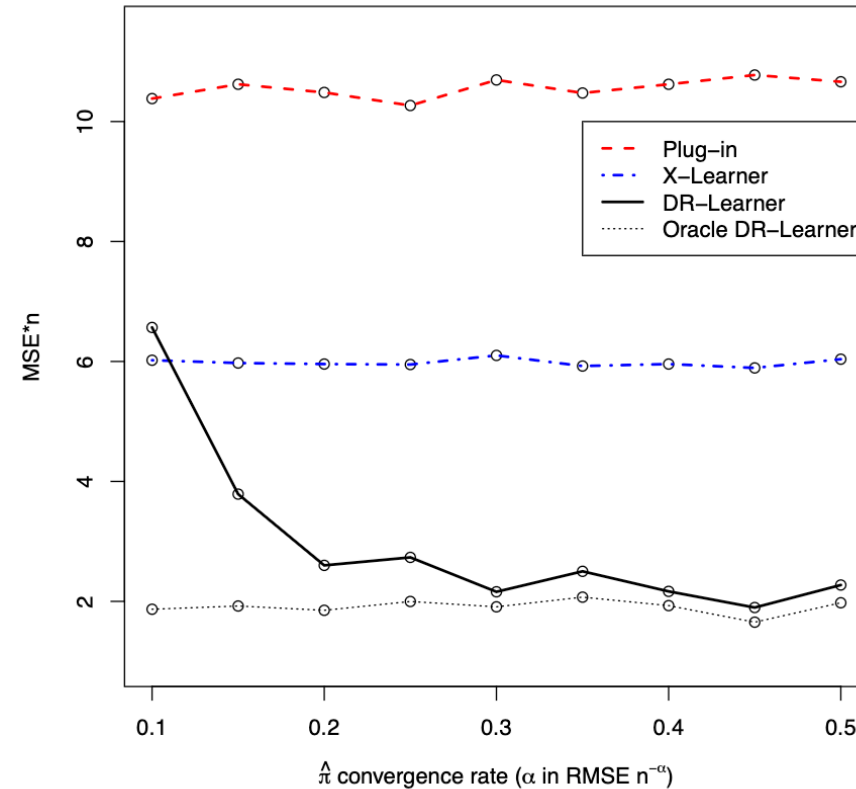


Kennedy's simulation: rough μ_a , flat τ .
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$$\hat{b}(x) = E[\hat{\varphi} - \varphi | X = x] = (e - \hat{e}) \left\{ \frac{\mu_1 - \hat{\mu}_1}{\hat{e}} + \frac{\mu_0 - \hat{\mu}_0}{1 - \hat{e}} \right\}$$

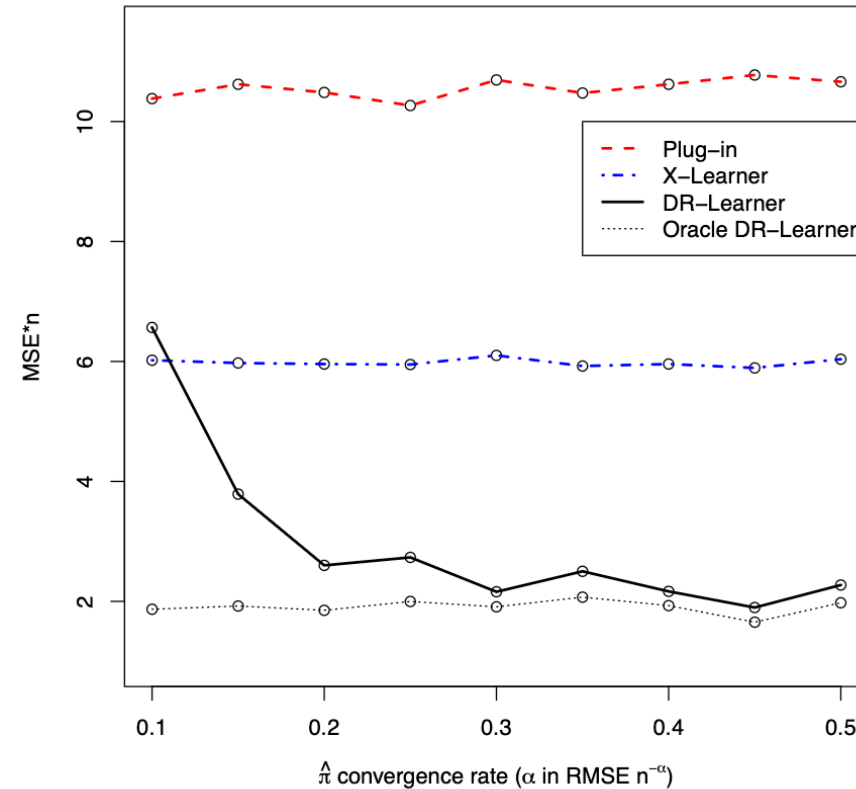
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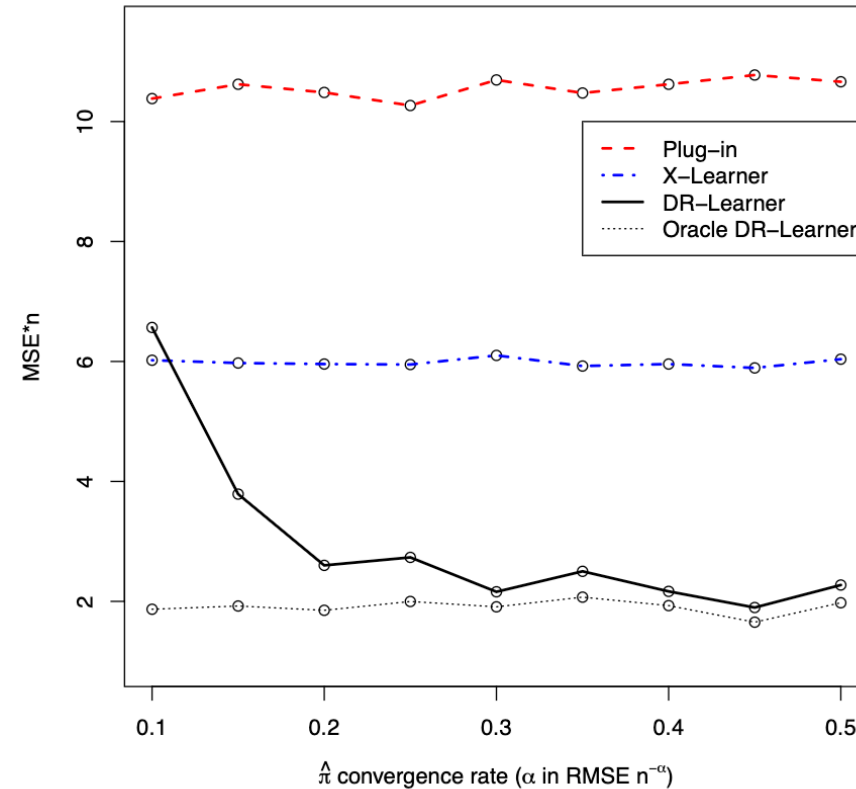
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- › We're using our outcome models as a **control variate** to reduce variance
 - » For a stratified second-stage model, *this is just AIPW*
- › **In practice:** pseudo-outcomes are noisy (they divide by $\hat{e}(1 - \hat{e})$), so clip $\hat{e}$ and regularise the second stage. In an RCT with known e , $\hat{b}(x) = 0$ however bad the $\hat{\mu}_a$ are.



References

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